# Report: Cardio Risk Prediction

## 1. Exploratory Data Analysis (EDA)

*Your report should present a clear and thorough EDA of the training data, including data cleaning steps, univariate and bivariate analyses, numerical feature distributions and outliers, correlation analysis, class balance checks, target likelihood analysis for categorical variables, and relationships between numerical features and the target.*

*Use appropriate visualisations with meaningful interpretations that link back to the business objective.*

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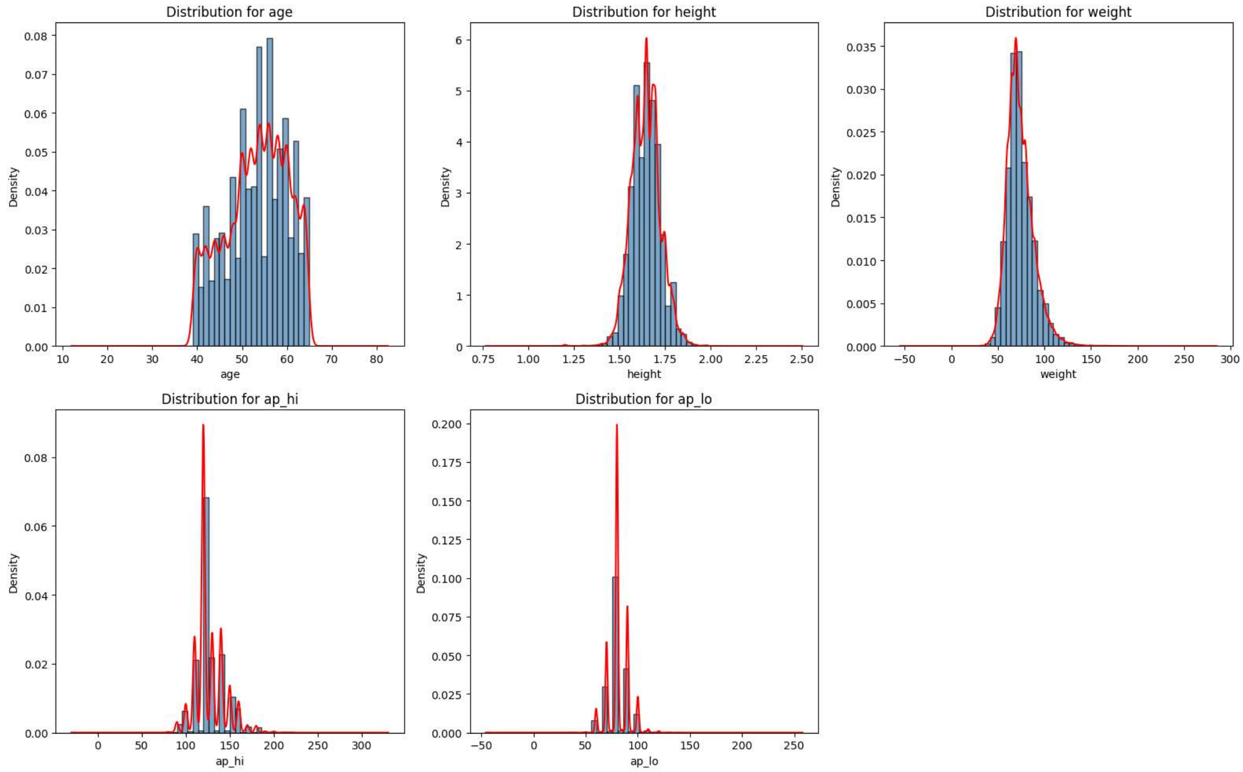
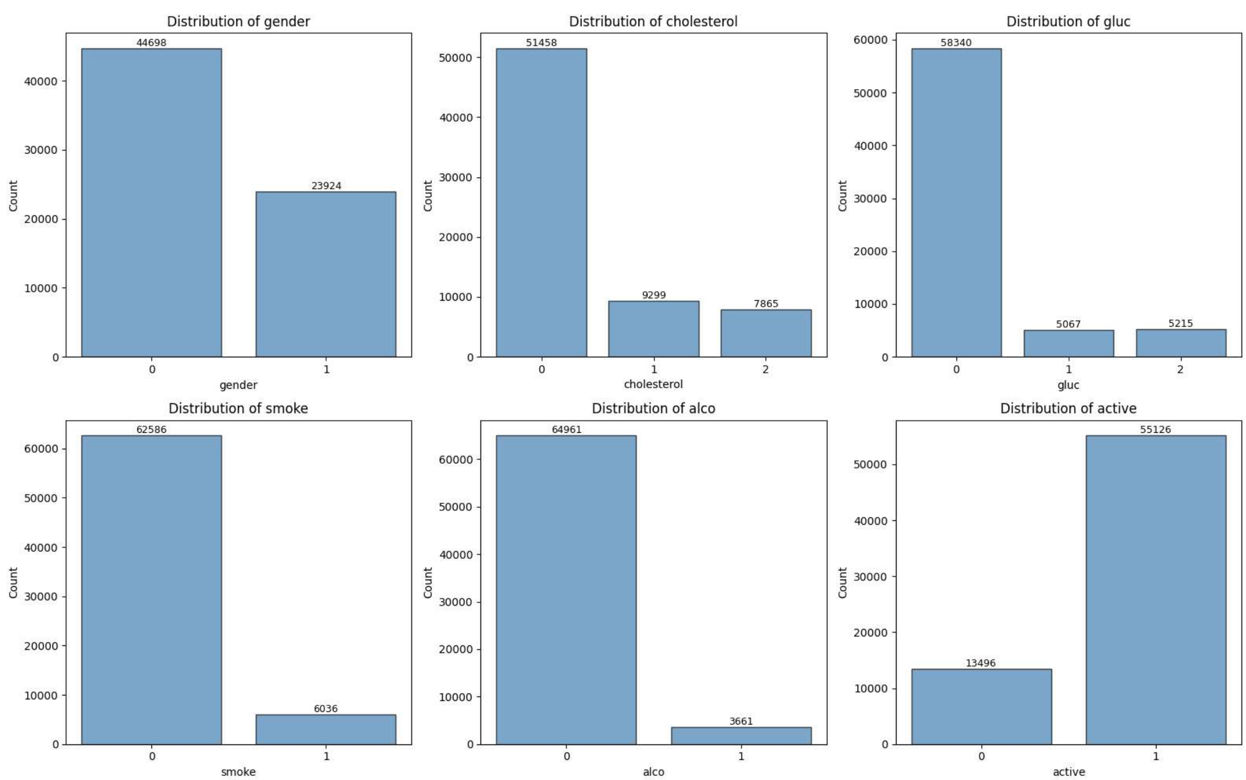
**Data Cleaning:**
The dataset contained 70,000 records with 14 columns. After removing irrelevant identifier columns (Unnamed: 0, id), the following cleaning steps were performed:

- Physiologically invalid rows were removed using clinical limits: systolic BP (ap_hi) outside [30, 300], diastolic BP (ap_lo) outside [30, 300], illogical BP where ap_lo >= ap_hi, height outside [120, 220] cm, and weight outside [30, 200] kg. This removed approximately 1-2% of records.
- Age was converted from days to years (dividing by 365.25) resulting in a range of ~29-65 years.
- Height was converted from centi-meters to meters for standard BMI calculation.
- Categorical columns (gender, cholesterol, gluc, smoke, alco, active, cardio) were converted to the category dtype. Cholesterol and glucose categories 1 and 2 were merged into a single "above normal" category as both showed similar CVD risk elevation.

**Univariate Analysis:**
- Age is approximately bell-shaped, centered around 50-55 years.
- Systolic (ap_hi) and diastolic (ap_lo) BP show sharp peaks at typical values (120/80 mmHg) with right-skewed tails.
- Height follows a roughly normal distribution; weight is slightly right-skewed.
- Lifestyle features (smoke, alco) are heavily skewed toward 0 — the majority of patients are non-smokers and non-drinkers.
- Physical activity (active) shows ~80% of patients are physically active.
- Cholesterol and glucose are predominantly at the normal level (0), with fewer patients in the elevated categories.

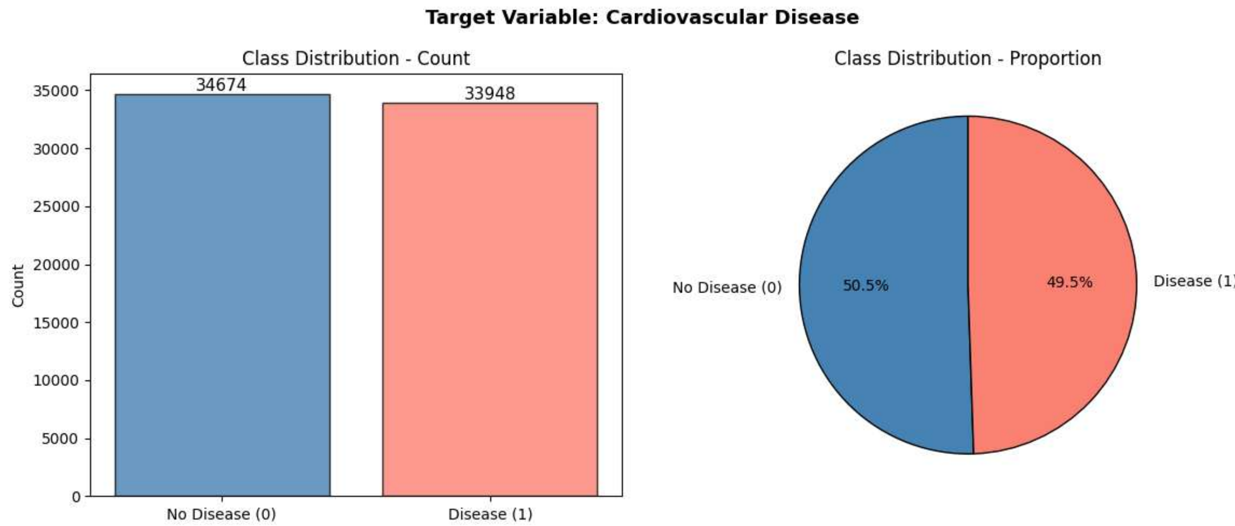



**Class Balance:**

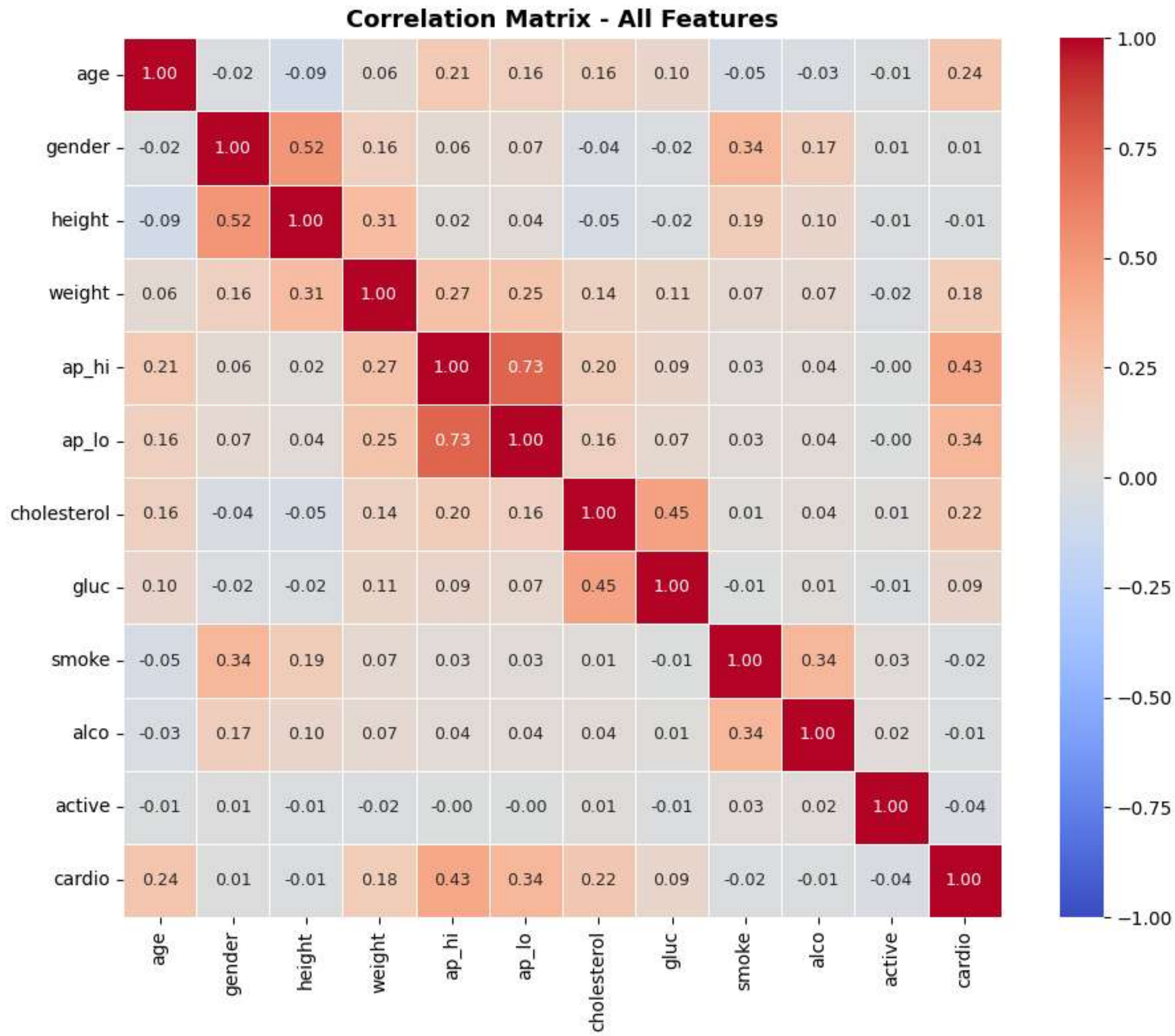
The target variable (cardio) is nearly balanced — approximately 50% No Disease and 50% Disease.

This is favorable for model building as accuracy remains a meaningful metric and no class weighting or oversampling is required.



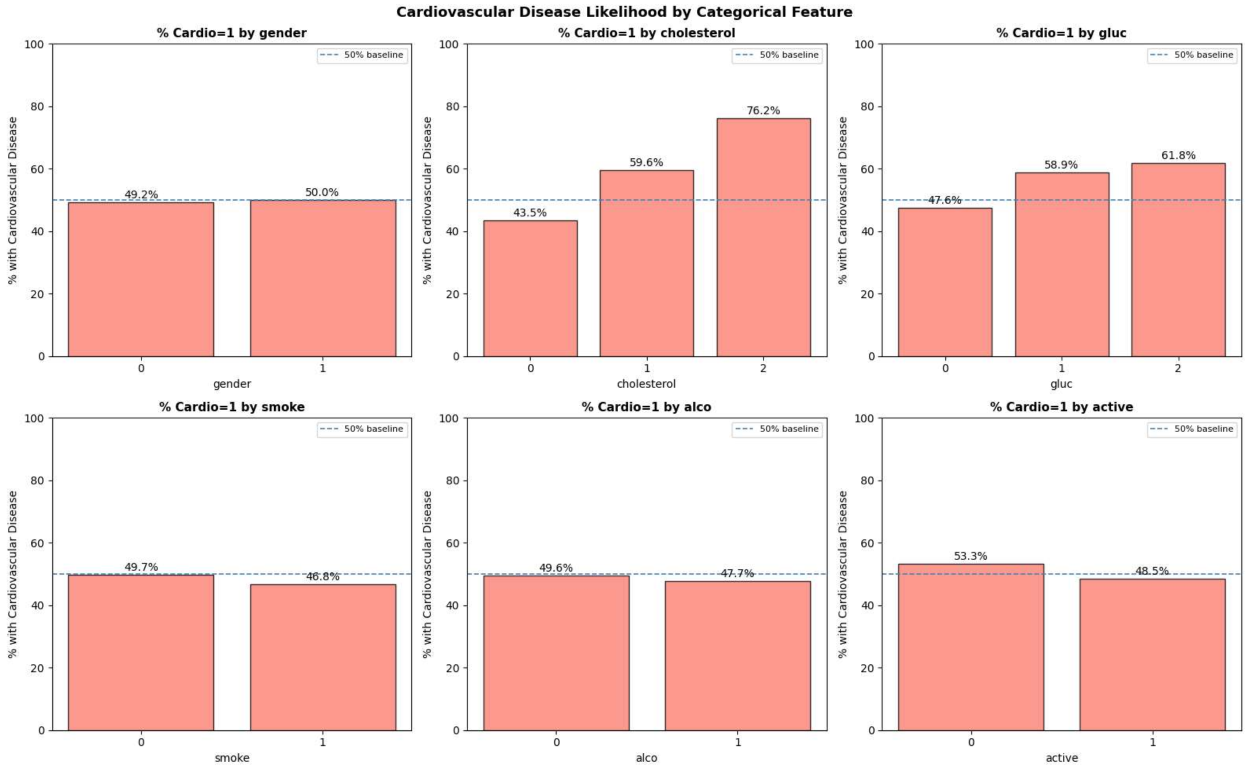

**Correlation Analysis:**

- ap_hi and ap_lo show moderate positive correlation (~0.5-0.6), as expected since systolic and diastolic BP tend to move together.
- **Age and ap_hi** show **the strongest positive** correlations with the target variable cardio.
- Height shows near-zero correlation with cardio, confirming it is a weak predictor.
- No feature pair shows correlation above 0.8, so multicollinearity is not a major concern among raw features.



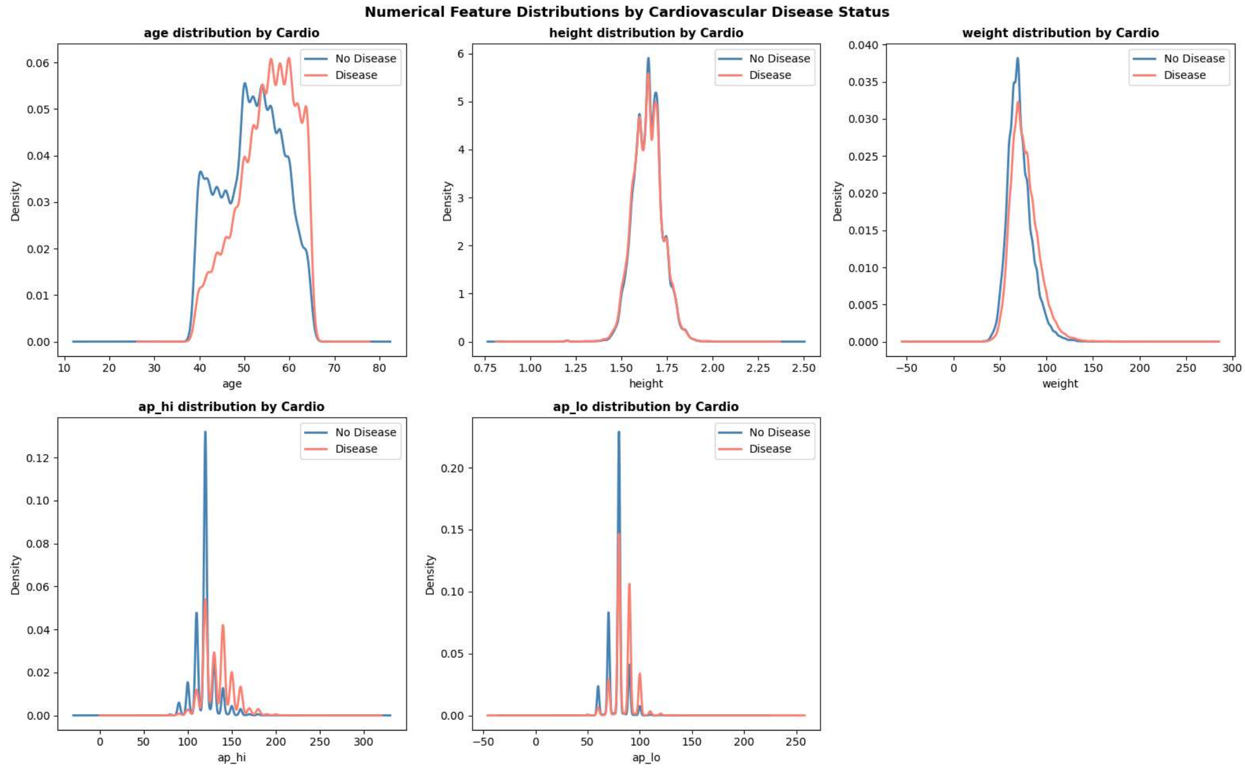

**Bivariate Analysis — Categorical Features:**

- Cholesterol analysis vs target shows that patients with elevated cholesterol (category 1) show a noticeably higher proportion of CVD cases compared to normal (category 0). This is one of the more informative categorical features.
- Glucose vs category 1 show similar pattern to cholesterol but weaker.
- Smoke, alco: CVD proportions are near 50% for both categories — these features offer limited discriminatory power.
- Active: physically active patients show slightly lower CVD risk, though the difference is modest.
- Gender: minimal difference in CVD proportion between gender categories.




**Bivariate Analysis — Numerical Features vs Target:**

- Age: the disease group's KDE curve is clearly shifted right — older patients have higher CVD risk. Strong predictor.
- ap_hi: disease group shows higher systolic BP readings on average. Strong predictor.
- ap_lo: similar shift to ap_hi but less pronounced. Moderate predictor.
- Weight: slight right-shift for disease group. Mild predictor.
- Height: distributions for both groups are almost identical. Very weak predictor.




## 2. Model Building and Rationale

*Your report should clearly describe all candidate models (e.g., Decision Tree, SVM, additional models if used), the rationale for selecting them, the feature selection or engineering process, the choice of cutoffs, and justification for hyperparameters. The report should explain the full model-building process and why each modelling decision is appropriate for the data and task.*

**Candidate Models:**
Three classification models were built and evaluated:

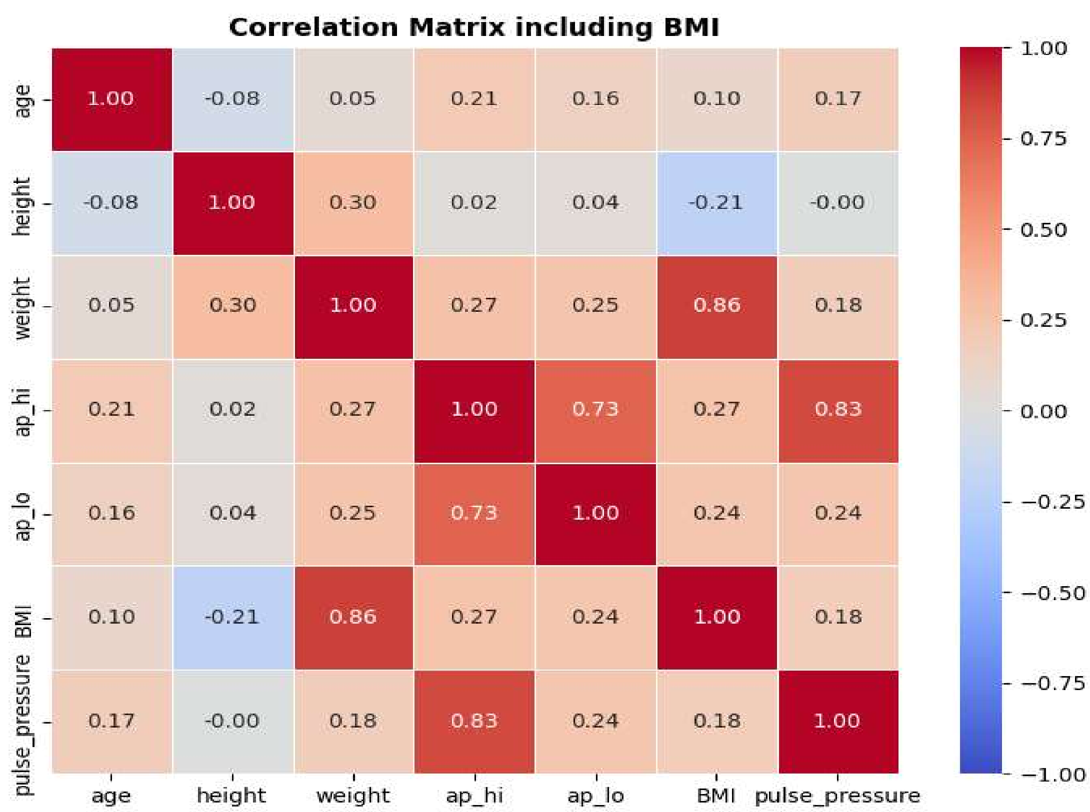
1. Linear SVM (Support Vector Classifier with linear kernel)
2. Decision Tree Classifier
3. Random Forest Classifier (ensemble of Decision Trees)

**Rationale for Model Selection:**

- **Linear SVM** was chosen as a strong generalizing linear model that works well when data is approximately linearly separable. It is widely used in medical classification tasks and provides calibrated probabilities via Platt scaling.
- **Decision Tree** was chosen for its interpretability and ability to generate feature importance scores, which directly support the business question of identifying key CVD risk factors.
- **Random Forest** was chosen as the third model — a natural ensemble extension of the Decision Tree that addresses overfitting through averaging across 100 trees. It typically achieves better generalization than a single decision tree.

**Feature Engineering:**

- BMI (Body Mass Index) = weight / height$^2$ was created as a standard clinical metric. Weight was dropped after confirming high correlation with BMI (r > 0.85) to avoid multicollinearity.
- Pulse Pressure = ap_hi - ap_lo was created as an independent cardiovascular risk indicator. It ranked 4th in Decision Tree feature importance, confirming its predictive value.
- Cholesterol and glucose levels were re-encoded from 3 categories to binary (normal vs above normal) based on EDA showing similar CVD risk between categories 1 and 2.
- All numerical features were scaled using MinMaxScaler fitted on training data only.
- Categorical features were dummy-encoded with drop_first=True to avoid the dummy variable trap.



**Cutoff Selection:**
- Default cutoff of 0.5 was evaluated first.
- Sensitivity-specificity curves were plotted to identify the intersection point as the balanced optimal threshold.
- Given the healthcare priority of minimising false negatives (missed CVD cases), the threshold was selected to prioritise recall over precision, accepting a higher false positive rate in exchange for higher sensitivity.

**Hyperparameter Tuning:**
- Decision Tree: GridSearchCV with 5-fold CV, scoring='recall'. Parameters tuned: max_depth [3,5,7,10,15], min_samples_split [10,20,50,100], min_samples_leaf [5,10,20,50], max_features ['sqrt','log2',None], criterion ['gini','entropy'].
- Random Forest: GridSearchCV with 5-fold CV, scoring='recall'. Parameters tuned: n_estimators [100,200], max_depth [5,10], max_features ['sqrt','log2'].
- SVM (optional): GridSearchCV on 20% stratified subset due to computational cost. Parameters tuned: kernel ['linear','rbf'], C [0.1,1,10], gamma ['scale','auto']. RBF SVM achieved AUC of 0.7908 vs 0.7917 for Linear SVM — confirming the linear boundary is appropriate.

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## 3. Model Cards

Your report should include complete model cards for all candidate models, covering overview, intended use, data and features used, model configuration, performance results, and limitations or considerations.

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**Model Card 1: Linear SVM**

Model overview:
A Support Vector Machine with a linear kernel trained to classify patients as at-risk or not at-risk for cardiovascular disease. Primary recommended model for deployment.

Intended use:
- Binary classification of cardiovascular disease risk (0: No Disease, 1: Disease)
- Intended for healthcare providers and clinical decision-support systems at CardioCare
- Suitable for triage workflows to prioritise patients for further consultation

Data and features:

- Raw numerical features: age (years), height (m), ap_hi, ap_lo
- Engineered features: BMI (weight/height$^2$), pulse_pressure (ap_hi − ap_lo)
- Dropped: weight (high correlation with BMI, r > 0.85)
- Dummy-encoded: gender_1, cholesterol_1, gluc_1, smoke_1, alco_1, active_1
- Numerical features scaled using MinMaxScaler (fit on training data only)

Model configuration:

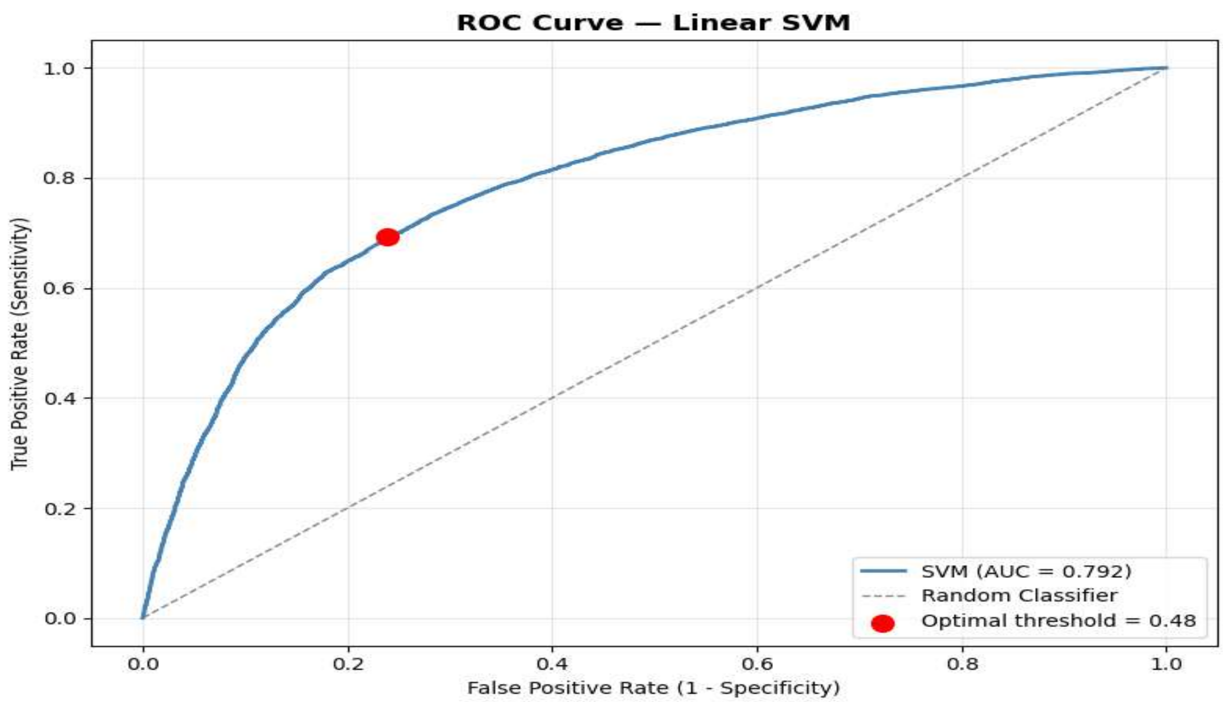
- Algorithm: SVC, Kernel: Linear, probability=True, random_state=42
- Threshold chosen from sensitivity-specificity intersection

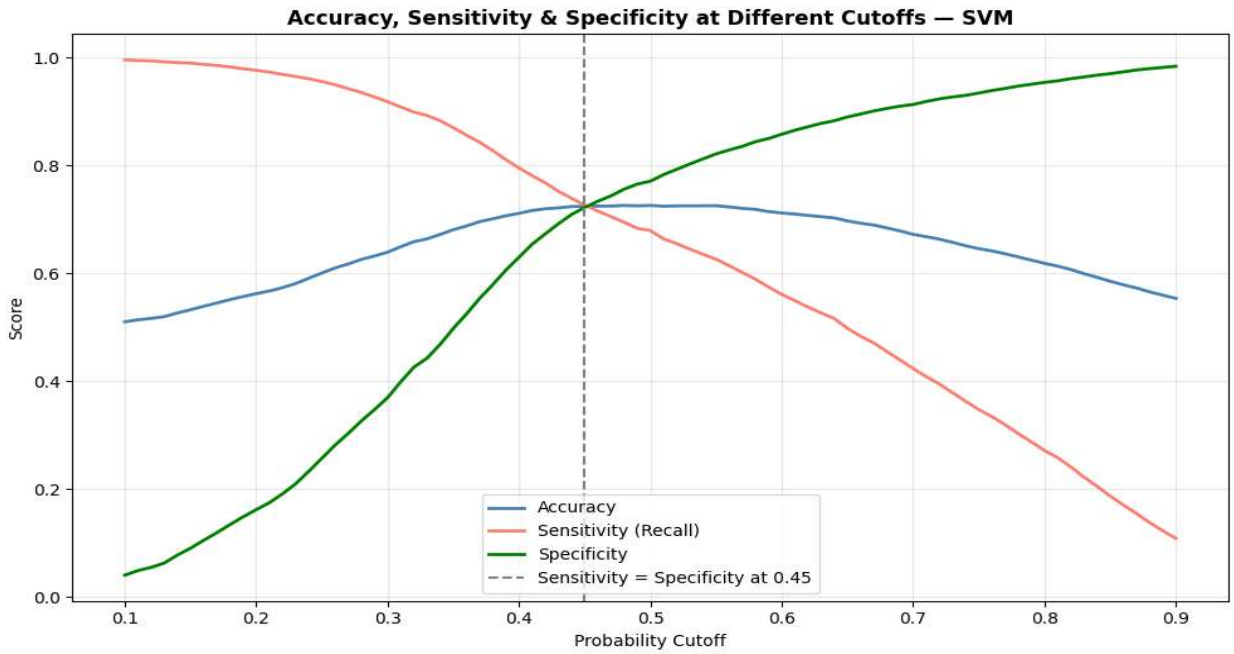
Performance:

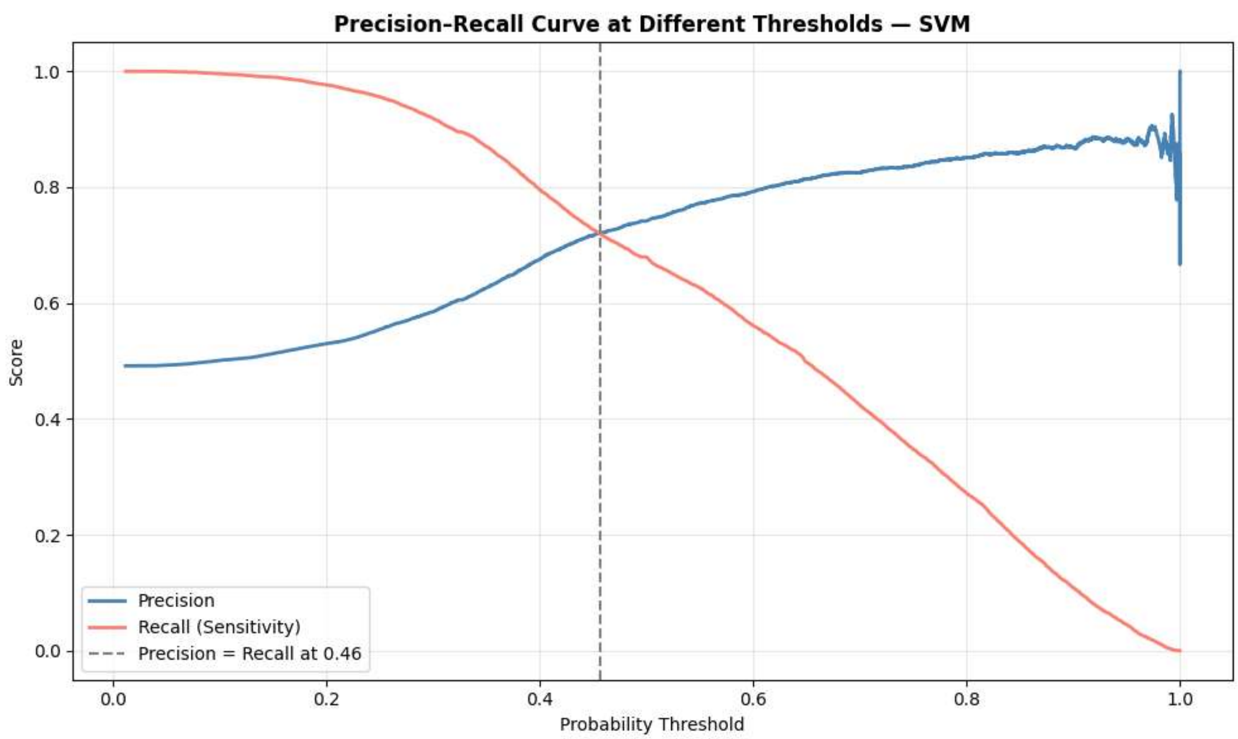
Train Accuracy: 0.7246 | Test Accuracy: 0.7245 | Precision: 0.7169 | Recall: 0.7267 | F1: 0.7218 | AUC: 0.7917 | Overfit Gap: 0.0000

Limitations:

- Linear boundary may miss complex non-linear interactions (though RBF SVM showed no improvement)
- Computationally expensive to train on large datasets
- Does not account for patient history or longitudinal trends



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**Model Card 2: Decision Tree (Baseline)**

Model overview:
An unconstrained Decision Tree used as a baseline. Not recommended for deployment due to severe overfitting.

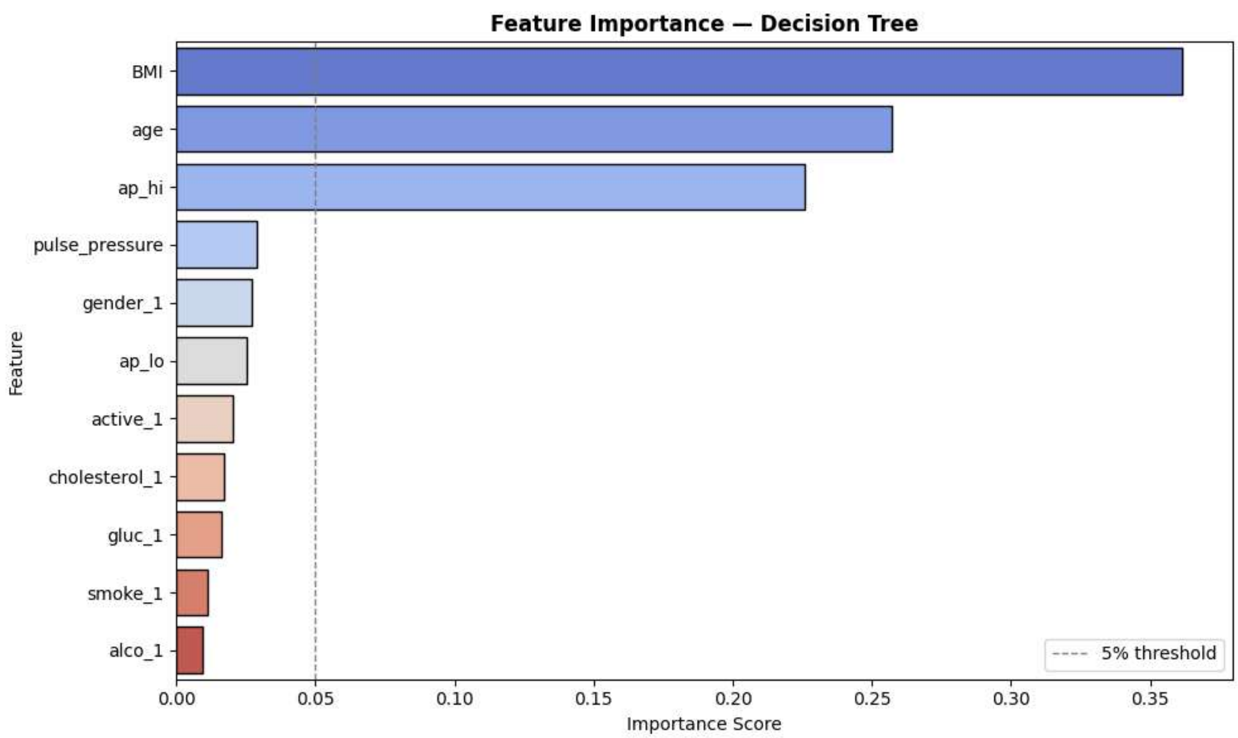
Intended use:
- Baseline comparison only; not suitable for clinical deployment

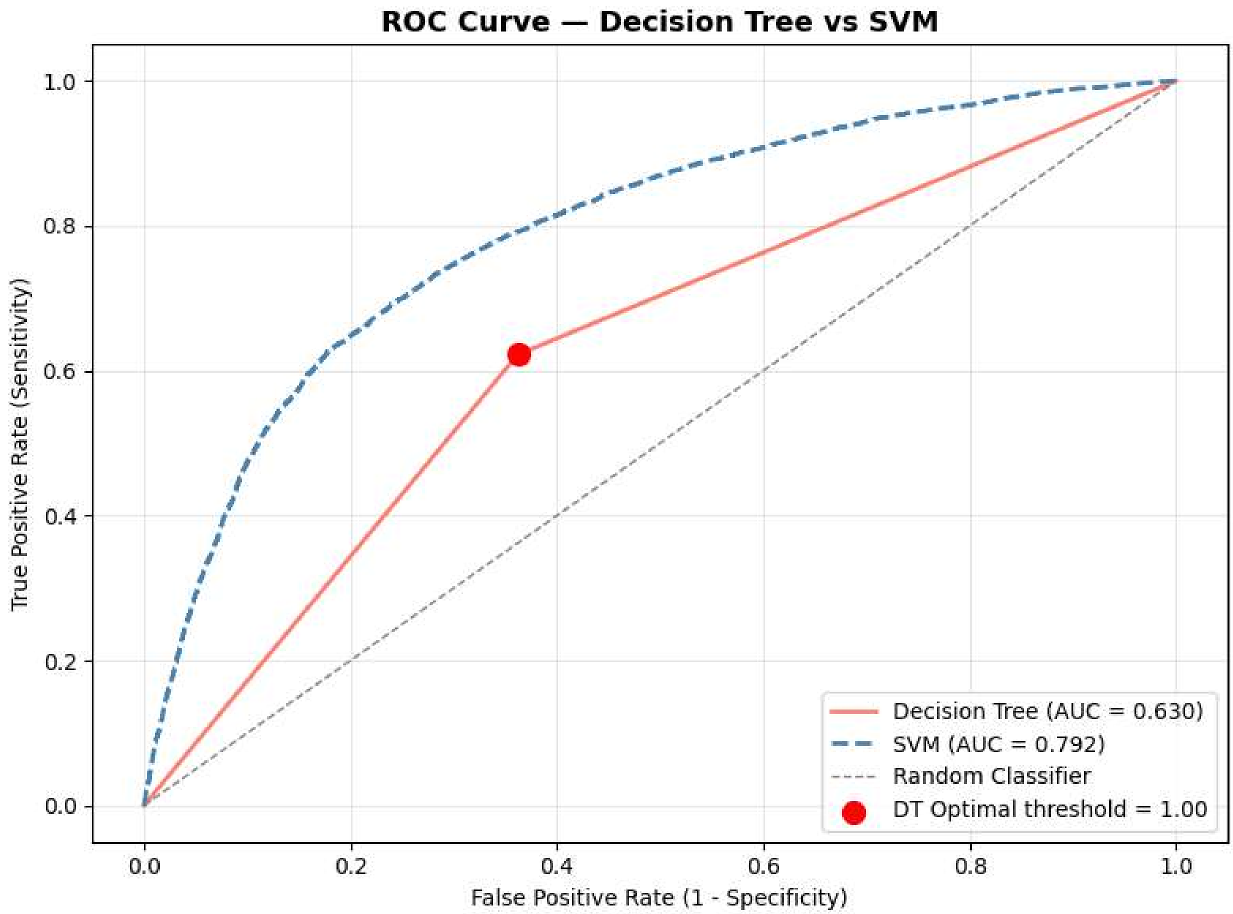
Data and features: Same as Model Card 1

Model configuration:
- Algorithm: DecisionTreeClassifier, no depth constraints, random_state=42

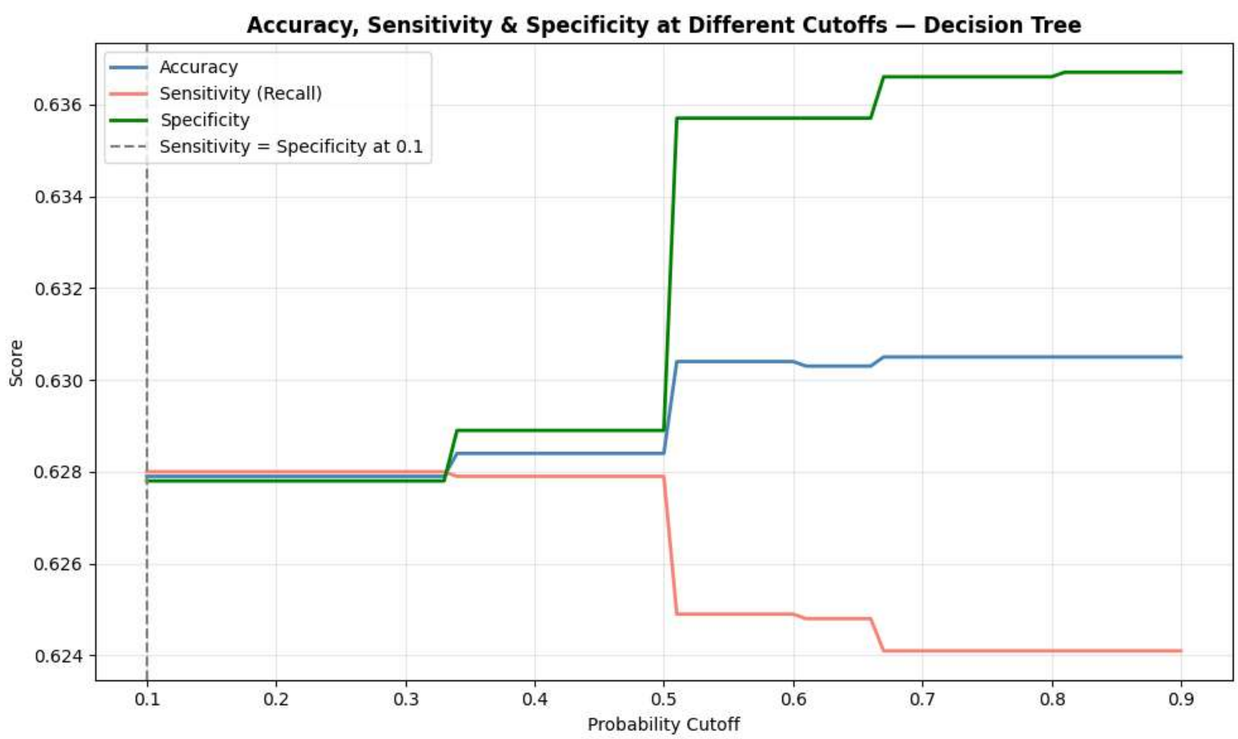
Performance:
Train Accuracy: 0.9954 | Test Accuracy: 0.6279 | Precision: 0.6200 | Recall: 0.6280 | F1: 0.6240 | AUC: 0.6300 | Overfit Gap: 0.3674

Limitations:
- Severe overfitting — not suitable for deployment
- Serves as baseline to demonstrate need for regularization

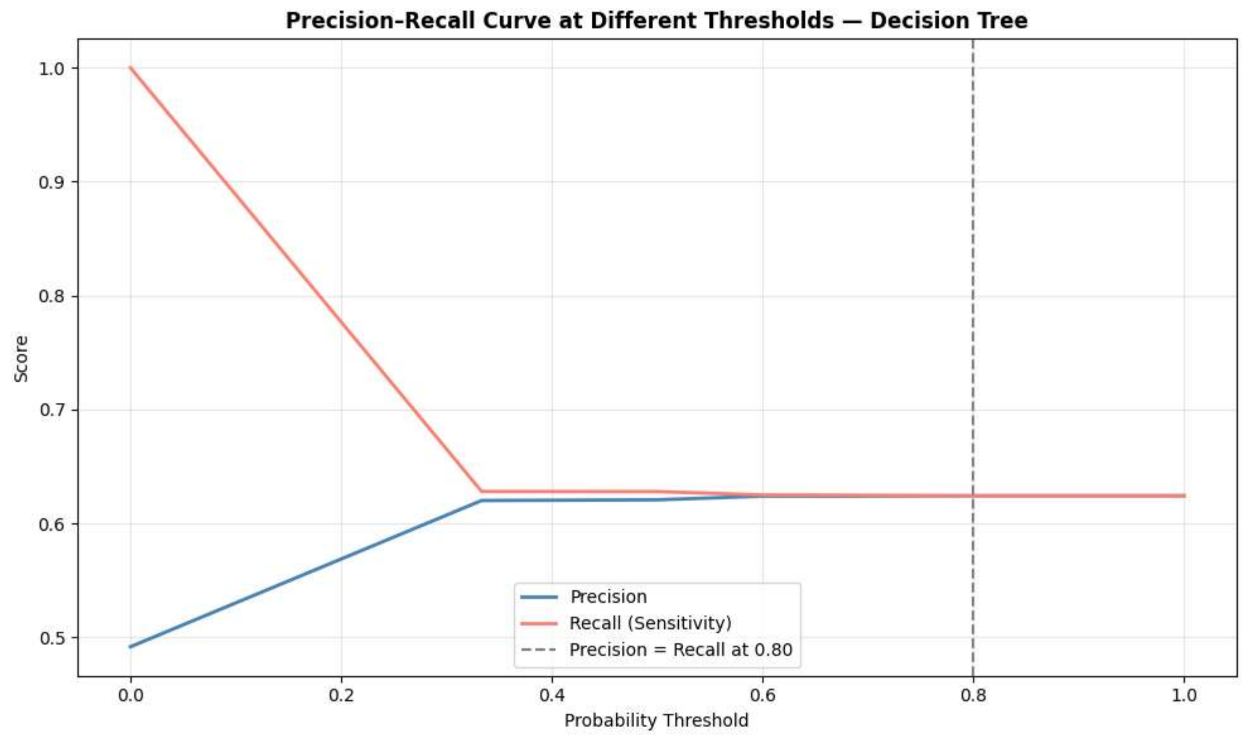




ROC Curve — Decision Tree vs SVM
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
Decision Tree (AUC = 0.630)
SVM (AUC = 0.792)
Random Classifier
DT Optimal threshold = 1.00

Accuracy, Sensitivity & Specificity at Different Cutoffs — Decision Tree
Accuracy
Sensitivity (Recall)
Specificity
Sensitivity = Specificity at 0.1
Score
Probability Cutoff




**Model Card 3: Tuned Decision Tree**

Model overview:
Decision Tree with hyperparameters optimised via GridSearchCV (recall scoring). Overfitting resolved but aggressive recall optimisation leads to very low precision.

Intended use:
- Experimental only; not recommended for balanced clinical use

Data and features: Same as Model Card 1

Model configuration:
- Algorithm: DecisionTreeClassifier, tuned via GridSearchCV, scoring='recall', 5-fold CV, random_state=42

Performance:
Train Accuracy: 0.5309 | Test Accuracy: 0.5262 | Precision: 0.5093 | Recall: 0.9932 | F1: 0.6733 | AUC: 0.7945 | Overfit Gap: 0.0048

Limitations:
- Flags nearly all patients as disease-positive (recall 0.99, precision 0.51)
- Overall accuracy of 52.6% is unacceptable for clinical use without threshold re-calibration

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**Model Card 4: Random Forest (Baseline)**

Model overview:
Ensemble of 100 unconstrained Decision Trees. Improves on single tree but still overfits significantly.

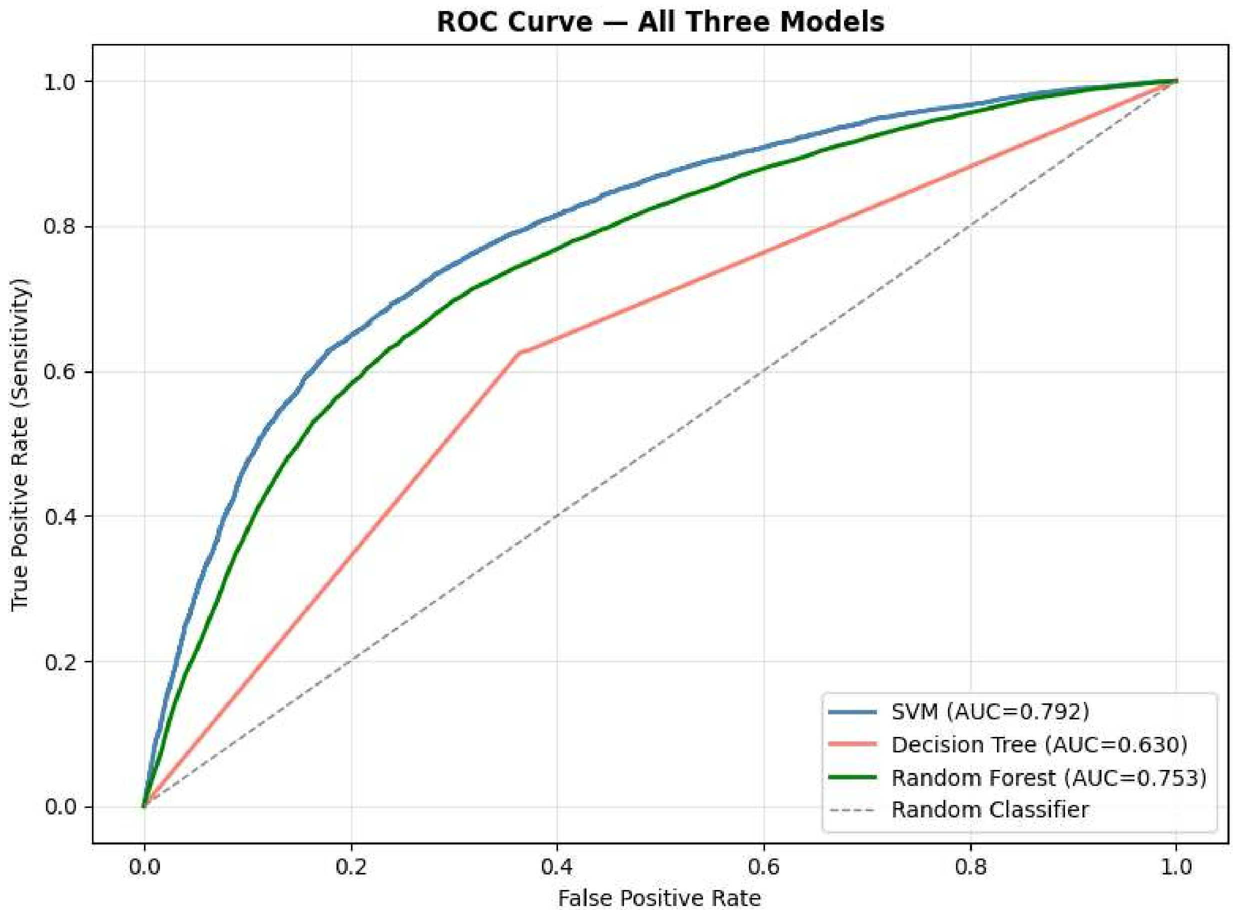
Intended use:
- Intermediate comparison model; not suitable for deployment without tuning

Data and features: Same as Model Card 1

Model configuration:
- Algorithm: RandomForestClassifier, n_estimators=100, no depth constraints, random_state=42, n_jobs=-1

Performance:
Train Accuracy: 0.9957 | Test Accuracy: 0.6986 | Precision: 0.6923 | Recall: 0.6966 | F1: 0.6944 | AUC: 0.7532 | Overfit Gap: 0.2970

Limitations:
- Significant overfitting (gap of 29.7%) — hyperparameter tuning required

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**Model Card 5: Tuned Random Forest**

Model overview:
Random Forest with hyperparameters optimised via GridSearchCV. Best overall AUC among all models with good generalisation.

Intended use:
- Suitable for deployment as an alternative to Linear SVM, particularly if AUC is the primary metric

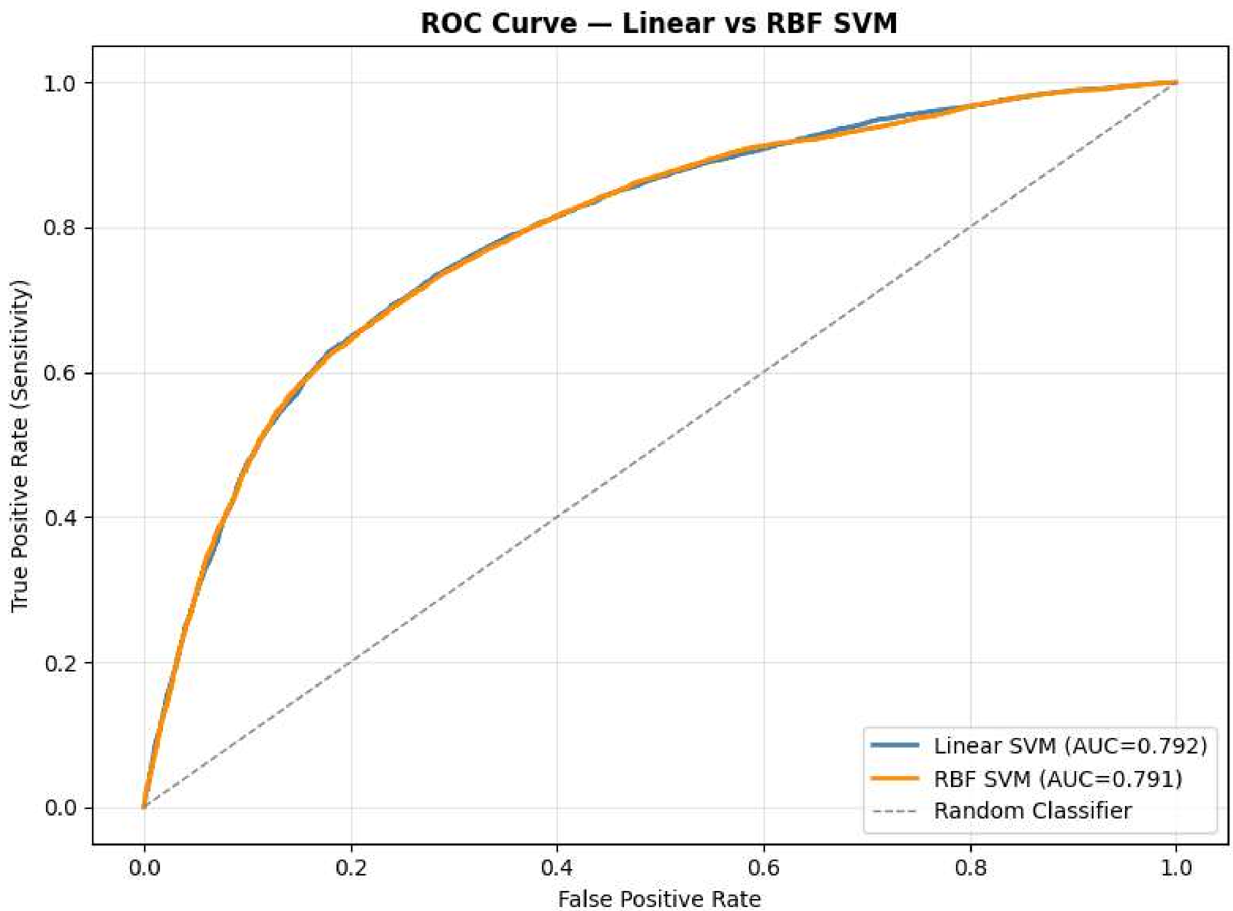
Data and features: Same as Model Card 1

Model configuration:
- Algorithm: RandomForestClassifier, tuned via GridSearchCV, scoring='recall', 5-fold CV, random_state=42, n_jobs=-1

Performance:
Train Accuracy: 0.7454 | Test Accuracy: 0.7289 | Precision: 0.7367 | Recall: 0.6982 | F1: 0.7169 |

AUC: 0.7990 | Overfit Gap: 0.0165

Limitations:
- Less interpretable than Linear SVM (ensemble of many trees)
- Marginally lower recall (0.698) than Linear SVM (0.727)
- Computationally heavier to train and tune

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## 4. Final Conclusion

Your report should summarise key insights from all stages: important findings from features and EDA, baseline vs tuned model performance, overfitting issues and how they were resolved, assessment of data sufficiency, the suitability of linear vs non-linear models, the reasoning behind the final model choice, and the final outcomes of the modelling process.

**Key EDA Insights:**
The most important predictors identified through EDA were age, systolic blood pressure (ap_hi), and BMI. These three features showed the clearest separation between CVD and non-CVD patient distributions. Lifestyle features (smoking, alcohol, physical activity) showed surprisingly weak individual predictive power, with CVD proportions near 50% for both categories. Cholesterol was the strongest categorical predictor. The target variable was nearly balanced (~50/50), which made accuracy a reliable evaluation metric throughout.

**Feature Engineering:**
Two engineered features were created — BMI (weight/height$^2$) and pulse pressure (ap_hi − ap_lo). Both proved valuable: BMI became the single most important feature in tree-based models (importance ~0.36), and pulse pressure ranked 4th overall. Weight was dropped due to high correlation with BMI ($r > 0.85$). Cholesterol and glucose categories were simplified from 3 to 2 levels based on EDA showing similar risk between elevated categories.

**Baseline vs Tuned Model Performance:**
Baseline models revealed a clear contrast between SVM and tree-based approaches. The Linear SVM generalised well from the start (test accuracy 72.5%, AUC 0.792, zero overfit gap). Both the Decision Tree and Random Forest baseline models overfitted severely — the Decision Tree had a train/test gap of 36.7% and AUC of only 0.63, while the Random Forest had a gap of 29.7%. After hyperparameter tuning, the Decision Tree's gap reduced to 0.5% and the Random Forest's gap reduced to 1.65% with AUC improving from 0.753 to 0.799.

**Overfitting — Detection and Resolution:**
Overfitting was clearly detected through the large discrepancy between training (~99.5%) and test (~62-70%) accuracy in both tree-based baseline models. This was resolved through GridSearchCV with 5-fold cross-validation, optimising for recall. Key hyperparameters tuned included max_depth, min_samples_split, min_samples_leaf, and max_features. Constraining tree depth was the most impactful change — it prevented the model from creating overly specific rules that only applied to training data.

**Data Sufficiency:**
The dataset with ~68,000 clean records (after removing physiologically invalid entries) was sufficient for this task. The balanced class distribution, consistent train/test performance for the Linear SVM, and stable cross-validation scores all confirm the data volume was adequate to learn reliable patterns without overfitting concerns for the SVM model.

**Linear vs Non-linear Model Suitability:**

A linear model is sufficient for this dataset. The RBF kernel SVM achieved an AUC of 0.7908 versus 0.7917 for the Linear SVM — a negligible difference of 0.001. This indicates the decision boundary in this data is largely linear. The engineered features (BMI, pulse pressure) likely contributed to this by transforming non-linear raw relationships into more linear signals.

**Final Model Choice and Outcomes:**

**Linear SVM is selected as the final recommended model** based on:

- Zero overfit gap (0.0000) — train and test accuracy are virtually identical at 72.5%
- Highest recall (0.727) among well-generalising models — critical for minimising missed CVD cases in a healthcare context
- Competitive AUC (0.7917), only 0.007 below the best model (Tuned Random Forest)
- Simplicity, fast inference, and transparency suitable for clinical deployment

**The Tuned Random Forest (AUC 0.799, gap 1.65%) is a strong alternative** if discriminative ranking is prioritised over sensitivity. Overall, the models demonstrate that cardiovascular disease risk can be predicted with approximately 72-73% accuracy and AUC of ~0.79-0.80 from basic health metrics — supporting CardioCare's goal of prioritising high-risk patients for clinical consultation.

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